



WINNER RANKINGS

1	Horseshoe False Discovery Rate Control via Frequentist Thresholding Im	65%
2	T-JEPA Tabular Masking with TF-Gene Interaction Structure Prevents Co- ◀	63%
3	Inverted JEPA Masking Aligns Pretext Directionality with TF-to-Gene Ca	62%
4	GMM-Anchored JEPA Pretraining Grounds W_init in Regulatory Logic Beyon	62%
5	Conditional Independence Masking in JEPA Targets Suppresses Co-express	60%

T-JEPA Tabular Masking with TF-Gene Interaction Structure Prevents Co-expression Collapse in Stage 1

Status: Winner · **Domain:** Self-Supervised Representation Learning / Gene Regulatory Network Inference · **Generation:** 0 · **Score:** 63%

Abstract

The Stage 1 weight predictor in two-stage GRN inference fails to learn causal regulatory signal because standard masking strategies in JEPA treat all input features equally, allowing the encoder to reconstruct expression profiles by exploiting highly correlated co-expressed gene clusters rather than regulatory relationships. T-JEPA demonstrates that augmentation-free self-supervised learning for tabular data succeeds when masking respects feature structure rather than applying random masks [51], yet no existing GRN method uses biologically structured masking—specifically, masking entire TF columns during prediction—to force the encoder to learn regulatory rather than co-expression patterns. We hypothesize that a structured TF-column masking strategy in Stage 1, where entire transcription factor expression profiles are masked and must be predicted from the remaining TFs and target genes, will force W_{init} to encode TF-specific regulatory influence rather than pairwise gene co-expression. This strategy makes regulatory prediction the pretext task itself, aligning Stage 1's optimization objective with Stage 2's goal. On DREAM5 net3, masking the full expression vector of randomly sampled TFs and requiring their reconstruction from other TFs creates a task that cannot be solved by co-expression alone, since TF-specific regulatory patterns are required. Expected outcome: W_{init} AUROC > 0.60 on net3, enabling Stage 2 AUPR > 0.099.

Paper [51] (T-JEPA) demonstrates that for tabular data, JEPA-style prediction works best when masking respects the feature structure of the domain, avoiding augmentations that destroy task-relevant correlations. Paper [30] identifies that co-expression profiling misrepresents biological interactions because noise and indirect effects dominate when all gene pairs are treated symmetrically. The gap is that no current JEPA-based GRN encoder uses TF-specific structural masking as a pretext task, despite the biological asymmetry between TFs (regulators) and target genes being the exact distinction Stage 1 must learn.

2. HYPOTHESIS STATEMENT

Replacing random feature masking in Stage 1 JEPA with structured TF-column masking (masking entire TF expression profiles and predicting them from remaining TFs and target genes) will force W_{init} to encode TF-specific regulatory influence, increasing $net3$ W_{init} AUROC from <0.58 to >0.60 .

3. BACKGROUND & LITERATURE REVIEW

- **T-JEPA: Augmentation-Free Self-Supervised Learning for Tabular Data** — Thimonier et al. (2024)
- **A Novel Graph Transformer Framework for Gene Regulatory Network Inference** — Teji et al. (2025)
- **DSeq-JEPA: Discriminative Sequential Joint-Embedding Predictive Architecture** — He et al. (2025)
- **Var-JEPA: A Variational Formulation of the Joint-Embedding Predictive Architecture -- Bridging Predictive and Generative Self-Supervised Learning** — Gögl et al. (2026)
- **Machine Learning Methods for Gene Regulatory Network Inference** — Hegde et al. (2025)
- **Soft Clustering Anchors for Self-Supervised Speech Representation Learning in Joint Embedding Prediction Architectures** — Ioannides et al. (2026)
- **Transformer-based feature extraction approach for hematopoietic cancer subtype classification.** — Park et al. (2026)
- **Inferring gene regulatory networks from single-cell data: a mechanistic approach** — Herbach et al. (2017)
- *+ 18 additional papers — see Section 8.*

4. METHODOLOGY

Modify Stage 1 JEPA to use TF-column masking: at each training step, randomly select k TFs ($k \sim 10\text{--}20\%$ of all TFs), zero out their expression values across all samples, and train the encoder to predict the masked TF expression profiles from the visible context (remaining TFs + all target genes). The weight matrix W_{init} is extracted from the encoder's learned mapping from target genes to TF predictions. Evaluate AUROC and Stage 2 AUPR.

Experimental Phases

1. Modify the Stage 1 data loader to implement TF-column masking: at each epoch, randomly select $k = \text{ceil}(0.15 \times n_{\text{TF}})$ transcription factors, replace their expression values with zeros across all samples, and define the prediction target as the masked TF expression profiles (not reconstructed in pixel space but in latent space following JEPA conventions).
2. Train Stage 1 for 150–300 epochs using the structured masking objective alongside a standard MSE loss; compare loss convergence to random-feature-masking baseline to confirm the structured masking task is harder (slower convergence expected).
3. Extract W_{init} from the encoder's final layer attention weights or linear projection mapping target gene space to TF prediction space; evaluate AUROC on held-out DREAM5 gold standard edges for each network.
4. Run ablation: (a) random gene masking (not TF-specific), (b) TF-column masking with $k=5\%$, $k=15\%$, $k=30\%$ to find optimal mask rate, (c) masking TF columns but using pixel-space (expression-space) reconstruction instead of latent-space prediction as in standard MAE — the last ablation distinguishes JEPA vs. MAE regimes for this masking strategy.
5. Pass W_{init} to Stage 2 horseshoe regression; record AUPR on all four networks; specifically analyze whether net3 AUPR exceeds GENIE3's 0.099 threshold.
6. Measure Stage 1 training time with TF-column masking vs. baseline to confirm $<3\text{s}$ per 10 epochs; TF-column masking does not add parameters, only changes the sampling pattern, so no slowdown is expected.

5. EXPECTED OUTCOMES

Predicted Impact: W_{init} AUROC increase from <0.58 to >0.60 on net3; net3 Stage 2 AUPR increase to >0.099 ; structured masking requires no additional parameters or compute, maintaining Apple Silicon MPS feasibility; potential benefit across all four networks as TF-specific regulatory patterns are universal to GRN inference.

Success Criteria: W_{init} AUROC > 0.60 on net3 with TF-column masking vs. < 0.58 with random masking (statistically significant at $p < 0.05$ over 5 random seeds); Stage 2 net3 AUPR > 0.099 ; net2 AUPR ≥ 0.0164 ; Stage 1 training time $< 3\text{s}$ per 10 epochs on Apple Silicon MPS; TF-column masking outperforms random masking on W_{init} AUROC on at least 3 of 4 networks.

Null Result Interpretation: If this hypothesis is false: Reverse the masking direction: mask target gene columns and train the encoder to predict target gene expression from visible TF

expression profiles only, which directly encodes TF→gene regulatory influence in the causal direction and produces a more naturally interpretable W_{init} .

6. RISK ANALYSIS & MITIGATION

Fatal Assumptions

- ☠ The direction of the masking prediction task (target genes predicting TFs) aligns with or is equivalent to learning TF regulatory influence on target genes — if false: the trained encoder captures gene-to-TF predictability (the reverse relationship), and W_{init} encodes inverse-regulatory signal that may perform worse than standard co-expression JEPA.
- ☠ A single interpretable weight matrix W_{init} can be cleanly extracted from the encoder's distributed representation of the target-gene-to-TF prediction — if false: the architectural modification required to produce a meaningful W_{init} may be non-trivial and arbitrary, invalidating the AUROC comparison.

Weakest Assumption: That training the encoder to predict masked TF columns from target gene expression (anti-causal direction) forces learning of TF→gene regulatory programs — this confuses the prediction task direction with the causal direction and would actually reinforce anti-causal representations.

Key Risks

- The proposed masking direction (predict masked TF columns from visible target genes) trains the encoder in the anti-causal direction; W_{init} extracted from the mapping 'target genes → TF predictions' captures how well target gene expression explains TF activity, not how TF activity drives target expression.
- With $k = 10\text{--}20\%$ of TFs masked per step, the remaining context still includes highly correlated TFs; the encoder can solve the reconstruction task by interpolating from co-expressed TFs without learning any target-gene-to-TF regulatory information, preserving the co-expression problem.
- Extracting W_{init} from 'the encoder's learned mapping from target genes to TF predictions' is architecturally ambiguous — in a standard JEPA transformer, this mapping is distributed across attention heads and is not naturally a single interpretable weight matrix without architectural constraints.
- *E. coli* (net3) has ~334 TFs; masking 10–20% means masking 33–67 TFs per step, which may be too aggressive for the encoder to converge reliably on the reconstruction task, leading to high training loss and uninformative representations.

Minimal Validation Experiment

Implement TF-column masking on net2 (smaller network) and compare the directionality of learned W_{init} associations: compute the asymmetry between $W_{init}[TF_i \rightarrow gene_j]$ and $W_{init}[gene_j \rightarrow TF_i]$ scores. Takes ~2 days. If W_{init} is symmetric (no directional preference), the masking has not imposed causal directionality and the hypothesis is falsified; if asymmetric in the correct causal direction, proceed to net3.

- $\Rightarrow$ Alternative 1 (Masking Rate Sensitivity): Any structured masking rate above ~5% will succeed equally well because the co-expression problem is driven by the loss function, not the masking pattern—if true, random masking with equivalent rates would achieve the same W_{init} AUROC as TF-column masking, implying the structure is irrelevant.
- $\Rightarrow$ Alternative 2 (MAE Reconstruction Equivalence): The latent-space prediction objective of JEPA is not necessary—predicting masked TF expression directly in expression space (MAE-style reconstruction) would achieve equivalent W_{init} AUROC because the TF-column structure, not the latent-space prediction, is the informative component.

Discriminating Experiment

Run Stage 1 with four conditions on net3: (A) TF-column masking with latent-space JEPA prediction, (B) random feature masking with latent-space JEPA prediction at matched rate, (C) TF-column masking with pixel-space MAE reconstruction, (D) random masking with MAE reconstruction. Measure W_{init} AUROC for all four. If THIS hypothesis is correct: $AUROC(A) > AUROC(B)$ and $AUROC(A) > AUROC(C)$ — both the TF-column structure AND the latent-space prediction contribute. If Alternative 1 is correct: $AUROC(A) \approx AUROC(B)$. If Alternative 2 is correct: $AUROC(C) \approx AUROC(A) > AUROC(B)$.

7. LIMITATIONS & ASSUMPTIONS

Key Assumptions (most $\rightarrow$ least confident)

1. Assumption [82%]: The Stage 1 encoder's weight matrix W_{init} can be extracted as a meaningful TF-to-gene mapping from the TF-column masking prediction head — Risk: depending on architecture, the latent prediction head may not decompose cleanly into a $TF \times gene$ weight matrix; may require an explicit linear readout layer.
2. Assumption [70%]: Masking entire TF columns is not too easy (solvable by co-expression of correlated TFs) and not too hard (requiring regulation knowledge unavailable in expression data) — Risk: if TFs are highly co-expressed with each other, column masking can be solved by copying neighboring TF profiles without learning any regulatory logic.
3. Assumption [58%]: The DREAM5 JEPA architecture has TF and gene expression as separate input channels that can be masked independently — Risk: if the current Stage 1 implementation treats all genes uniformly without TF/non-TF distinction, TF-column masking requires a non-trivial architectural refactor.
4. Assumption [45%]: W_{init} encoding TF-specific regulatory patterns (from the column masking task) transfers to better edge ranking in Stage 2 — Risk: Stage 2 horseshoe regression may not benefit from TF-specific initialization if its objective (edge-level regression) operates at a different granularity than the TF-level pretext task.

Major Assumptions — require revision if false

- $\triangle$ Masking TF columns provides a harder reconstruction task than random feature masking, forcing the encoder to use genuine cross-type (gene-TF) regulatory structure — if false: TF-

TF co-expression among the remaining visible TFs provides an easier reconstruction shortcut.

- $\triangle$ The masking rate $k = 10\text{--}20\%$ is in the right range for net3; too low leaves too much TF context available (shortcut), too high makes the task unlearnable — if false: hyperparameter search per network is needed, significantly increasing computational cost.

8. EVIDENCE SUMMARY

DIMENSION	SCORE	WEIGHT
EVIDENCE GROUNDING	55%	35%
Feasibility	68%	30%
Impact Potential	67%	25%
Novelty	65%	10%
Composite	63%	weighted

Why this won: This hypothesis is distinct from H1 and H2 because it modifies the masking structure of the pretext task itself rather than adding auxiliary loss terms or post-processing. It directly addresses the co-expression problem by making TF-specific regulatory prediction the training objective, exploiting the biological asymmetry between TFs and target genes. The T-JEPA paper [51] provides the key insight that structured masking outperforms random masking for tabular genomic data, and the approach requires zero additional parameters, making it trivially feasible on MPS. The discriminating experiment is clean because mask pattern is the only variable changed.

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